

MAGNETISM AND MATTER

Important formulas and concepts-

- “We can’t isolate the north or south poles of a magnet, if a bar magnet is cut into two halves then we get two similar bar magnets with somewhat weaker properties”.

❖ magnetic moment –

$$M = m l$$

- it is a vector quantity
- direction from south to north pole.

Here, m = pole strength

- Unit -Ampere-metre
 - It is directly proportional to area
- M is directly proportional to volume.

❖ Cutting of bar magnet-

- 1) If parallel to length, cut into ‘ n ’ equal parts

Then, pole strength, $m' = m/n$
Mag. Moment, $M' = M/n$

- 2) If perpendicular to length, cut into ‘ n ’ equal parts

Then, pole strength remains same
Mag. Moment, $M' = M/n$ (length $l' = l/n$)

❖ Magnetic field due to bar magnet(dipole)-

- 1) At axial point, $B = 2kM/r^3$ ($k = \mu_0/4\pi$)

- 2) At equatorial point, $B = kM/r^3$
(direction opposite to magnetic moment)

- 3) At any general point, $B = kM(3\cos^2\theta + 1)^{1/2} /r^3$

❖ **Properties of magnetic field lines –**

- **Direction of mag field lines**
 - **Outside magnet – north to south**
 - **Inside magnet - south to north**
- **Mag field lines of a magnet forms closed loops.**
- **The tangent to field line at any point gives direction of net magnetic field at that point.**
- **The larger the no. of field lines per unit area, the stronger is the magnitude of magnetic field.**
- **The magnetic field lines do not intersect as electric field lines.**

❖ **If magnet is cut into ‘n’ equal parts,**

- **Perpendicular to length, Time period $T' = T/n$**
- **Parallel to length, time period remains same**

❖ **Magnetic permeability-**

- **Ratio of magnitude of total magnetic field to that of magnetic intensity(H)**
$$\mu = B/H$$
- **The physical significance is that it measures the extent to which a magnetising field can penetrate a given magnetic material.**
- **Relative permeability,**

$$\mu_r = \mu_m / \mu_0$$

❖ **Magnetisation(M)-**

- **Net magnetic moment per unit volume.**

❖ Magnetic intensity(H)-

$$B = \mu_0 ni = \mu_0 H$$

$$(H = ni)$$

❖ Magnetic susceptibility-

➤ $\chi_m = M/H$

➤ a material with high value of χ_m can easily be magnetised.

❖ Relation b/w μ_r and χ_m -

$$\mu_r = 1 + \chi_m$$

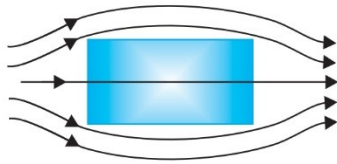
○ $B_{net} = B_o + B_m$
 $B_{net} = \mu_0 H + \mu_0 M$
 $B_{net} = \mu_0 (H + M)$
 $B_{net} = \mu_0 H(1 + \chi_m)$
 $B_{net} = \mu_0 H \mu_r$
 $B_{net} = \mu_m H$

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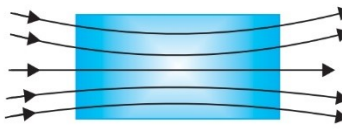
❖ Properties of magnetic materials-

Diamagnetic	Paramagnetic	Ferromagnetic
- Electrons are paired - Repel in external mag. Field - move from stronger to weaker part of mag. Field - magnetic susceptibility (χ_m) is negative and constant - net magnetic moment, $M_{net} = 0$ for both atom and crystal $\mu_m < \mu_0$ $B_m < B_0$ (B_m is mag. Field of medium i.e. diamagnetic material) (B_0 is mag field of free space) $\mu_r = 1 + \chi_m$ $\chi_m = -1$ $\mu_r = 0$ - examples are bismuth, copper, lead, silicon, nitrogen, water and sodium chloride	- electrons are unpaired - weakly attract in external mag. Field - weakly move from weaker to stronger part of magnetic field χ_m is small and positive $M_{net} \neq 0$ for atom but $M_{net} = 0$ for crystal due to random orientation of atom $\mu_m > \mu_0$ $B_m > B_0$ (B_m is mag. Field of medium i.e. paramag.) $\chi_m > 1$ $\mu_r > 1$ χ_m and μ_r depend not only on nature of material but also on temperature - examples are aluminium, sodium, calcium, oxygen and copper chloride - the individual atoms of paramag. Material possess a	- electrons are unpaired - strongly attract in external mag. Field - strongly move from weaker to stronger part of magnetic field χ_m is large and positive - same as paramagnetic $\mu_m \gg \mu_0$ $B_m \gg B_0$ (B_m is mag. Field of medium i.e. ferromag.) $\chi_m \gg 1$ $\mu_r \gg 1$ - examples are iron, cobalt, nickel, gadolinium, etc. - in some ferromagnetic material, magnetisation

○ diamagnetism is present in all the substances ○ most exotic diamagnetic materials are superconductors which exhibits both 'perfect conductivity' and 'perfect diamagnetism' this phenomenon of perfect diamagnetism is known as "Meissner Effect"	permanent magnetic dipole moment of their own. ○ Magnetisation is inversely proportional to temperature $M = CB_0 / T$ Equivalently, $\chi = C\mu_0 / T$ this is known as 'curie law' C is curie's constant	persists , such are known as '<u>hard ferromag. Materials</u>' such as Alnico, an alloy of iron, aluminium, cobalt, nickel and copper these materials form "permanent magnets" that can be used in things such as 'compass needle'. In other ferromagnetic materials, magnetisation disappears on removal of external field , such are known as '<u>soft ferro magnetic material</u>' such as 'soft iron'. ○ Above a certain temperature, ferromagnet becomes paramagnet Such temp is known as curie temperature T_c.
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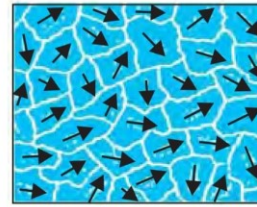


(a)

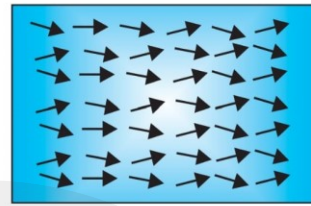


(b)

FIGURE 5.12
Behaviour of
magnetic field lines
near a
(a) diamagnetic,
(b) paramagnetic
substance.



(a)



B_0

(b)

FIGURE 5.13
(a) Randomly
oriented domains,
(b) Aligned domains.

SUMMARY

1. The science of magnetism is old. It has been known since ancient times that magnetic materials tend to point in the north-south direction; like magnetic poles repel and unlike ones attract; and cutting a bar magnet in two leads to two smaller magnets. Magnetic poles cannot be isolated.
2. When a bar magnet of dipole moment $\mathbf{m}$ is placed in a uniform magnetic field $\mathbf{B}$,

- (a) the force on it is zero,
- (b) the torque on it is $\mathbf{m} \times \mathbf{B}$,
- (c) its potential energy is $-\mathbf{m} \cdot \mathbf{B}$, where we choose the zero of energy at the orientation when $\mathbf{m}$ is perpendicular to $\mathbf{B}$.

3. Consider a bar magnet of size l and magnetic moment $\mathbf{m}$, at a distance r from its mid-point, where $r \gg l$, the magnetic field $\mathbf{B}$ due to this bar is,

$$\mathbf{B} = \frac{\mu_0 \mathbf{m}}{2\pi r^3} \quad (\text{along axis})$$

$$= -\frac{\mu_0 \mathbf{m}}{4\pi r^3} \quad (\text{along equator})$$

4. Gauss's law for magnetism states that the net magnetic flux through any closed surface is zero

$$\phi_B = \sum_{\text{all area elements } \Delta \mathbf{S}} \mathbf{B} \cdot \Delta \mathbf{S} = 0$$

5. Consider a material placed in an external magnetic field $\mathbf{B}_0$. The magnetic intensity is defined as,

$$\mathbf{H} = \frac{\mathbf{B}_0}{\mu_0}$$

The magnetisation $\mathbf{M}$ of the material is its dipole moment per unit volume. The magnetic field $\mathbf{B}$ in the material is,

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$$

6. For a linear material $\mathbf{M} = \chi \mathbf{H}$. So that $\mathbf{B} = \mu \mathbf{H}$ and χ is called the magnetic susceptibility of the material. The three quantities, χ , the relative magnetic permeability μ_r , and the magnetic permeability μ are related as follows:

$$\mu = \mu_0 \mu_r$$

$$\mu_r = 1 + \chi$$

7. Magnetic materials are broadly classified as: diamagnetic, paramagnetic, and ferromagnetic. For diamagnetic materials χ is negative and small and for paramagnetic materials it is positive and small. Ferromagnetic materials have large χ and are characterised by non-linear relation between $\mathbf{B}$ and $\mathbf{H}$.
8. Substances, which at room temperature, retain their ferromagnetic property for a long period of time are called permanent magnets.

POINTS TO PONDER

1. A satisfactory understanding of magnetic phenomenon in terms of moving charges/currents was arrived at after 1800 AD. But technological exploitation of the directional properties of magnets predates this scientific understanding by two thousand years. Thus, scientific understanding is not a necessary condition for engineering applications. Ideally, science and engineering go hand-in-hand, one leading and assisting the other in tandem.
 2. Magnetic monopoles do not exist. If you slice a magnet in half, you get two smaller magnets. On the other hand, isolated positive and negative charges exist. There exists a smallest unit of charge, for example, the electronic charge with value $|e| = 1.6 \times 10^{-19}$ C. All other charges are integral multiples of this smallest unit charge. In other words, charge is quantised. We do not know why magnetic monopoles do not exist or why electric charge is quantised.
 3. A consequence of the fact that magnetic monopoles do not exist is that the magnetic field lines are continuous and form closed loops. In contrast, the electrostatic lines of force begin on a positive charge and terminate on the negative charge (or fade out at infinity).
 4. A miniscule difference in the value of χ , the magnetic susceptibility, yields radically different behaviour: diamagnetic versus paramagnetic. For diamagnetic materials $\chi = -10^{-5}$ whereas $\chi = +10^{-5}$ for paramagnetic materials.
 5. There exists a *perfect diamagnet*, namely, a superconductor. This is a metal at very low temperatures. In this case $\chi = -1$, $\mu_r = 0$, $\mu = 0$. The external magnetic field is totally expelled. Interestingly, this material is also a perfect conductor. However, there exists no classical theory which ties these two properties together. A quantum-mechanical theory by Bardeen, Cooper, and Schrieffer (BCS theory) explains these effects. The BCS theory was proposed in 1957 and was eventually recognised by a Nobel Prize in physics in 1970.
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6. Diamagnetism is universal. It is present in all materials. But it is weak and hard to detect if the substance is para- or ferromagnetic.
 7. We have classified materials as diamagnetic, paramagnetic, and ferromagnetic. However, there exist additional types of magnetic material such as ferrimagnetic, anti-ferromagnetic, spin glass, etc. with properties which are exotic and mysterious.